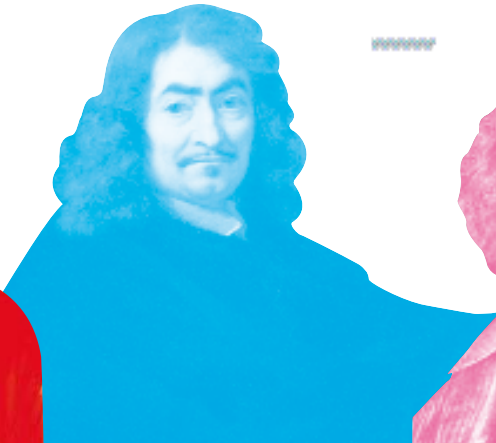




The machinery of
the heavens is not
like an animal
but like a clock

I think,
therefore
I am

The more I see of
men, the more I
like my dog



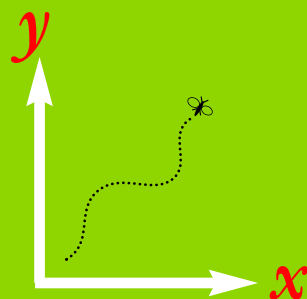
KEPLER
1571–1630

The German astronomer Johann Kepler measured the paths of the planets before telescopes were invented and found they orbited the Sun in ellipses, not circles. He showed that comets increase in speed as they near the Sun, and he found that a line drawn between the Sun and a planet will sweep over equal areas in equal times as the planet moves through its orbit.



DESCARTES
1596–1650

René Descartes watched a fly while lying in bed and thought “How can I explain the position of the fly at any moment?” He realized he could use three coordinates (x , y , and z) for each dimension of space (forward/back, up/down, left/right). Descartes was also the first person to use letters from the end of the alphabet to stand for values in algebra.



FERMAT
1601–1665

Pierre de Fermat created the most famous puzzle in maths – “Fermat’s last theorem”. He wrote in the margin of a book that he had found a “truly marvellous proof” that the equation $x^n + y^n = z^n$ cannot be solved if n is more than 2, but said “there is not enough room to write it here”. It took over 300 years to prove the theorem is true, but it seems likely that Fermat was lying.

$$x^n + y^n = z^n$$

PASCAL
1623–1662

A child prodigy, Pascal wrote a maths book at age 16 and invented a calculating machine made of cogs and wheels when he was 19. He worked on gambling puzzles with Fermat and in doing so founded probability theory and uncovered the patterns in Pascal’s triangle. At age 31 he became deeply religious and gave up maths to spend his last years in prayer and meditation.

